Sentiment Analysis Report: Amazon Product Reviews

Task: NLP Sentiment Analysis & Semantic Similarity | **Pipeline:** spaCy (en_core_web_md) + TextBlob

1. Description of the Dataset

The dataset utilized for this project is the comprehensive *Datafiniti Amazon Consumer Reviews of Amazon Products* dataset (May 2019 edition). It comprises **28,332 customer review entries** spanning 24 data fields, documenting consumer experiences across a wide assortment of Amazon products (including electronics, batteries, household essentials, and smart devices). The primary target feature for NLP modeling is **reviews.text**, representing unstructured text reviews. Supplementary columns include **reviews.rating** (1-5 star ratings), **reviews.doRecommend**, **reviews.title**, and **name** (product name).

2. Details of the Preprocessing Steps

Consumer review text presents natural irregularities including varied casing, colloquial formatting, punctuation noise, and non-informative stopwords. The preprocessing pipeline performs the following steps:

- **Missing Value Removal:** Applied dataframe.dropna(subset=['reviews.text']) to guarantee no null entries enter the NLP pipeline.
- **Case Standardization & Whitespace Trimming:** Transformed strings using str().lower().strip() to normalize lexical variance.
- **spaCy Token Filtering:** Processed text using en_core_web_md, filtering tokens via .is_stop, .is_punct, and .is_space attributes.
- **Cleaned Text Reconstruction:** Assembled filtered content tokens for downstream polarity and subjectivity evaluation.

3. Evaluation of Results

The sentiment model computes polarity scores ranging from **-1.0 (strongly negative)** to **+1.0 (strongly positive)**, and subjectivity scores from **0.0 (objective)** to **1.0 (subjective)**. Testing across diverse reviews demonstrated strong alignment between predicted polarity and recorded customer star ratings:

Sample Review Excerpt	Rating	Polarity	Predicted Sentiment
'...one of the item is bad quality. Is missing backup spring...'	3 / 5	-0.4500	Negative
'Thankful that I was able to find on Amazon for a great price...'	5 / 5	+0.6417	Positive
'Battery storage life only lasted 8 months (stored indoors).'	1 / 5	0.0000	Neutral (Objective)
'...buying AAA alkaline batteries... thought I was getting a pretty good deal...'	5 / 5	+0.3179	Positive
'Prior reviews were very good... So far so good! Highly satisfied...'	5 / 5	+0.4000	Positive

Semantic Similarity Evaluation: Using spaCy's 300-dimensional word vectors via similarity(), semantic similarity comparisons yielded meaningful contextual relationship scores: comparing Review 0 with Review 1 produced a similarity score of **0.9519**, and Review 0 with Review 15 produced a score of **0.9496**, demonstrating strong domain and syntactic clustering across product feedback.

4. Model Strengths and Limitations

Strengths:

- **Computational Efficiency:** Fast, lightweight execution capable of processing thousands of reviews without high GPU overhead.
- **Semantic Depth:** Pre-trained vectors in en_core_web_md capture domain relationships for semantic comparison beyond keyword matching.
- **Modular Architecture:** Easily extensible functions for batch pipelines, ETL jobs, and automated reporting.

Limitations:

- **Factual Complaints vs Sentiment:** Objective negative complaints (e.g., 'battery lasted 8 months') lack explicit emotional adjectives and can score neutral polarity (0.0000).
- **Complex Negation & Sarcasm:** Lexicon-based polarity calculation can miss multi-clause inversions or sarcastic phrasing compared to transformer architectures (e.g., RoBERTa/BERT).